

Exercises on Array Antennas, Part 2

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31 de julio de 2026

1. Compute by direct integration the radiation pattern of a half-wavelength dipole lying along $\hat{z}$ and centered at the origin. Assume that the z-directed current is given by $I(z) = I_0 \cos(kz)$, $|z| < \lambda/4$.

$$I(z) = I_0 \cos(kz) = I_0 \sin\left(\frac{\pi}{2} - kz\right) = I_0 \sin\left(\frac{2\pi}{\lambda}\left(\frac{\lambda}{4} - |z|\right)\right)$$

Como la antena es de media onda se tiene que $l = \frac{\lambda}{2}$

$$I = I_0 \sin\left(k\left(\frac{l}{2} - |z|\right)\right)$$

Sabemos que campo eléctrico es de la forma:

$$E_{far}(R) = jnkI_{in} \frac{e^{-jkr}}{4\pi r} l_e(\hat{r}) = \frac{e^{-jkr}}{4\pi r} e(\theta, \phi)$$

En forma integral se puede expresar de manera normalizada como:

$$\begin{aligned} e(\hat{r}) &= -jkn \int_{-l/2}^{l/2} e^{jkz' \cos\theta} \hat{r} \times (\hat{z} I_0 \sin(k(\frac{l}{2} - |z'|))) \times \hat{r} dz' \\ e(\hat{r}) &= \hat{\theta} kn I_0 \sin\theta \int_{-l/2}^{l/2} \frac{e^{jk(z' \cos\theta + l/2 - |z'|)} - e^{jk(z' \cos\theta + l/2 - |z'|)}}{2} dz' \\ e(\hat{r}) &= \hat{r} j 2n I_0 \frac{\cos(\frac{kl}{2} \cos\theta) - \cos(\frac{kl}{2})}{\sin\theta} \end{aligned}$$

Que es el patrón de radiación:

$$e(\theta) = \frac{\cos(\frac{\pi}{2} \cos\theta) - \cos(\frac{\pi}{2})}{\sin\theta} = \boxed{\frac{\cos(\frac{\pi}{2} \cos\theta)}{\sin\theta}}$$

2. It is required to design an antenna for broadcasting in horizontal polarization that has almost omni-directional coverage in the horizontal plane and directive pattern in the vertical plane.
 - a) Design a broadside array with minimum complexity (i.e. minimum number of elements) having $FNBW \leq 15^\circ$ and avoiding grating lobes at 850 MHz. Use a computer to plot the array factor in a relevant plane to check conformance to specs.

$$\begin{aligned} \frac{d}{\lambda} &= 1 - 1/N \\ \Psi < \frac{2\pi}{N} \rightarrow 2\pi \frac{d}{\lambda} + \cos\left(\frac{FNBW}{2} - \frac{\pi}{2}\right) < \frac{2\pi}{N} \\ \frac{1}{N \sin(\frac{FNBW}{2})} &< \frac{d}{\lambda} < 1 - 1/N \end{aligned}$$

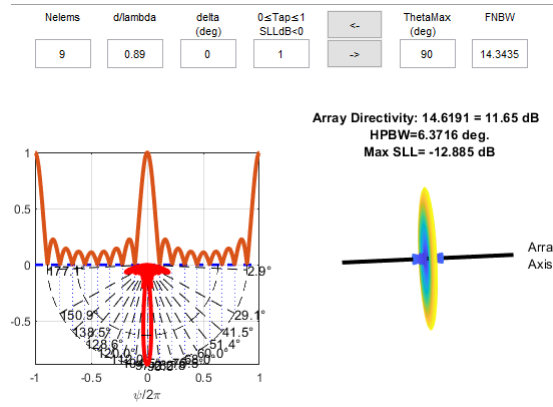
Para tener la menor complejidad posible (N_{min}) se debe cumplir:

$$\frac{1}{N \sin(\frac{FNBW}{2})} = 1 - 1/N$$

$$N = \frac{1}{\sin(\frac{FNBW}{2})} + 1$$

Y escogiendo el mayor ancho de haz posible (15°), para que tenga una mejor cobertura, se tiene que $N = 8,66 = 9$.

$$\frac{d}{\lambda} \approx 0,89$$



$$H_{far} = \frac{1}{n_0} \hat{r} \times E(r) \approx \hat{\phi} \frac{V_0}{n_0 r} \sin^{3/2} \theta \sum_n C_n e^{j k r_n \cdot \hat{r}} = \hat{\phi} \frac{V_0}{n_0 r} \sin^{3/2} \theta \frac{|\sin(\frac{N\Psi}{2})|}{|\sin(\frac{\Psi}{2})|}$$

iii. Compute the expressions for the field on these planes and use a computer to plot them.

Siguiendo el patrón de radiación en campo lejano se digito el siguiente código:

```
% parameters
f = 850E6;
N = 9;
lambda = 3E8 / f;
k = 2 * pi / lambda;
V0 = 1;

% Condition of Broadcast ULA
d = 0.89 * lambda;
delta = 0;

[theta, r] = ndgrid(linspace(0,2*pi,100),linspace(0,10,100));
Psi = k * d * cos(theta) + delta;
AF = abs(sin(Psi * N / 2)) ./ abs(sin(Psi / 2));
E = real((V0 ./ r) .* (sin(theta)).^1.5);
Efar = E .* AF;

% Convert spherical coordinates to Cartesian
[X, Y, Z] = sph2cart(theta,linspace(0,2*pi,100),r);

% Plot the 3D surface

surf(X, Y, Z,Efar);
title('ULA Broadside Antenna Pattern')
xlabel('x')
ylabel('y')
zlabel('z')
```

Figura 3: Código para generación de patrón

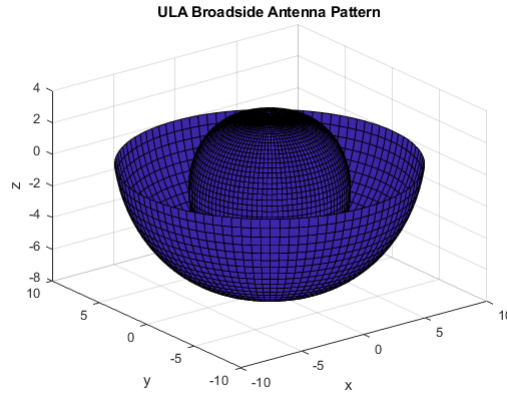


Figura 4: Patrón de radiación de arreglo incluyendo patrón de antena

El campo eléctrico se alinea en el plano azimutal, mientras que el campo magnético se orienta en el plano de elevación.

iv. Evaluate the frequency performance of the array designed. Considering a $\pm 5\%$ bandwidth, are the FNBW and GL goals achieved with your design? Re-design the array if required. Se ve que para el patrón en 2D que el ancho del haz FNBW es $97,39 - 82,96 = 14,43^\circ < 15^\circ$ y la ganancia máxima del lóbulo principal es de -0.21dB .

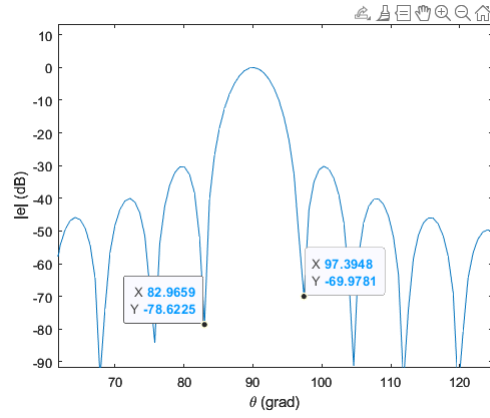


Figura 5: Patrón de radiación de arreglo incluyendo patrón de antena

Considerando una tolerancia del haz del $\pm 5\%$ *bandwidth*, es decir 5% de 15° , se tiene que el rango va de $16,68^\circ$ hasta $15,18^\circ$, por lo que se pasa de la restricción impuesta. Por lo tanto se vuelve a diseñar con el siguiente *FNBW*:

$$FNBW = 14^\circ \rightarrow N = 10, \frac{d}{\lambda} = 0,9$$

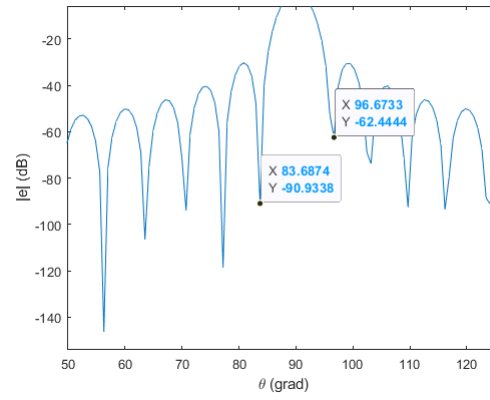


Figura 6: Patrón de radiación de arreglo incluyendo patrón de antena

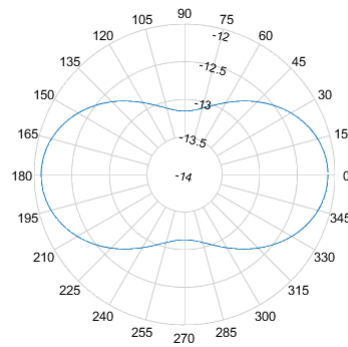


Figura 7: Antenna Array Bandwidth

Ahora $FNBW = 12,986^\circ$, y el rango está entre $12,24^\circ$ y $13,74^\circ$, sin embargo la ganancia o intensidad máxima del lóbulo principal disminuye a -0.26dB , es decir un $1,15\%$ en magnitud real, que no es muy significativo.

3. a) Design a rectangular patch antenna at 1850MHz having $R_{in} = 50\Omega$ using inset feed, i.e. compute L , W and l . Substrate parameters are $h = 1/20$ inch, $\epsilon_r = 2,33$. Se utilizaron las siguientes ecuaciones en matlab:

```
f=1850E6;
Zin=50;
er=2.33;
h=0.00127 ;
c=3E8;
lambda_0=c/f;
er_av=(er+1)/2;
Wopt=0.5*lambda_0/(sqrt(er_av));
er_eff=er_av+(er-1)/(2*sqrt(1+12*(h/Wopt)));
deltaL=h*0.412*(er_eff+0.3)/(er_eff-0.258)*((Wopt/h + 0.264)/(Wopt/h + 0.8));
lambda_g=lambda_0/sqrt(er_eff);
L=lambda_g/2 - 2*deltaL
```

Figura 8: Ecuaciones para el diseño de parche rectangular

$$L = 0,0526m, W_{opt} = 0,0628m$$

Como se está utilizando un inset feed, entonces se procede a encontrar L_g :

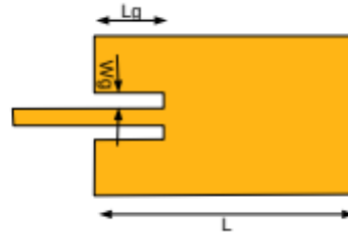


Figura 9: Parametros L_g y W_g de parche rectangular

$$R_{in} = R_{in}(L_g = 0) \cos^2\left(\frac{L_g \pi}{L}\right)$$

$$R_{in}(L_g = 0) = \frac{1}{G_{12} + G_e}$$

De las gráficas se obtiene que G_{12} y G_e a W_{opt} . Con estos valores, es posible encontrar la longitud L_g :

$$L_g = \sqrt{\frac{R_{in}}{R_{in}(L_g = 0)}} \cdot L/\pi$$

$$L_g = \sqrt{\frac{R_{in}}{R_{in}(L_g = 0)}} \cdot 0,0167$$

- b) Compute the frequency response of $|S_{11}|_{\text{dB}}$ by using the transmission line model with reference impedance $R_0 = R_{in}$ (patch input impedance at resonance) in a band $\pm 10\%$ around the resonant frequency above. What is the 10dB-bandwidth (in percent)?

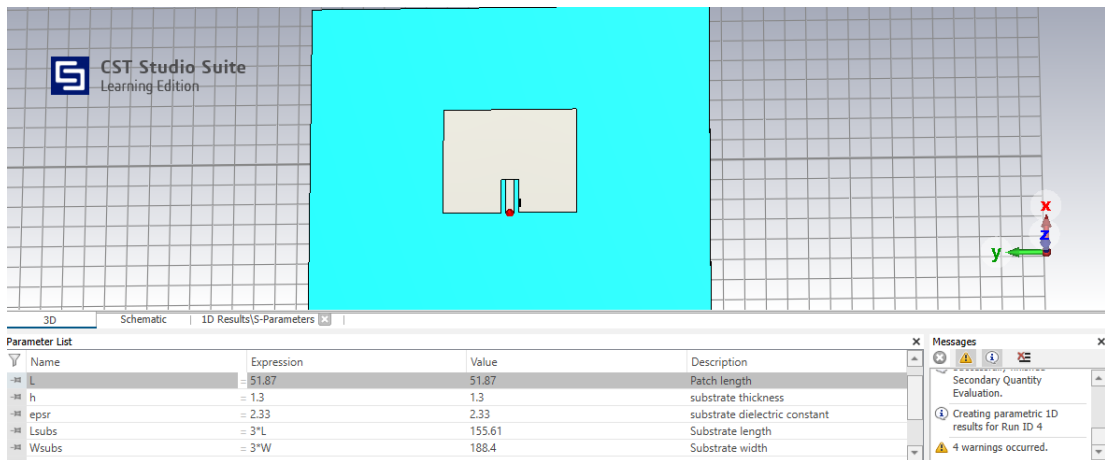


Figura 10: Diseño antenna tipo parche

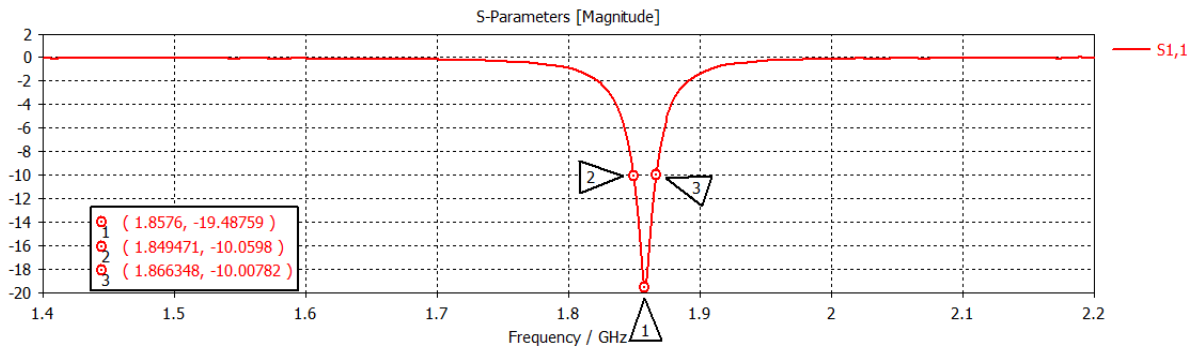


Figura 11: Frecuencia de resonancia antenna tipo parche diseñada

El porcentaje de ancho de banda a 10dB es:

$$BW \% = \frac{1,866 - 1,849}{1,849} * 100 = 0,92 \%$$

- c) Compute and plot the resulting pattern when the patch designed is used as basic element in the array of the previous exercise.

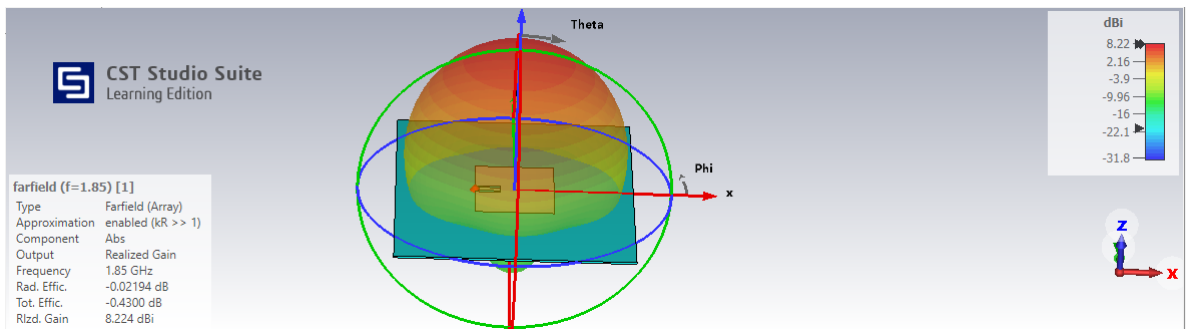


Figura 12: Patron de radiacion Antena tipo parche

4. For a $\lambda/2$ dipole placed symmetrically along the z-axis, determine dipole's total length.

a) Vector effective height l_e

$$AF = \frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta}$$

$$l_e = \frac{2}{AF} = \frac{2 \sin(\theta)}{\cos(\frac{\pi}{2} \cos \theta)}$$

b) Maximum magnitude of l_e A partir de ecuaciones de la densidad de potencia incidente de la antena, apertura eficiente, y el teorema de máxima transferencia de potencia que se ven en el libro Balanis, se llega a que:

$$A_e = \frac{n |l_e|^2}{4 R_{rad}}$$

$$R_{rad} \approx 20 \pi^2 (1/2)^2$$

$$A_{emax} \approx 0,13 \lambda^2$$

Por lo tanto :

$$l_{emax} = \sqrt{\frac{4 A_e R_{rad}}{n}} = 5,06 \frac{\lambda}{\sqrt{n}}$$

c) Maximum open-circuit voltage when a uniform plane wave with broadside incidence on the dipole and electric field $E = \hat{\theta} \cdot 10^{-3} \frac{V}{\lambda}$ The open circuit voltage of the circuit model is:

$$V = Z_{in}(E^i l_e)$$

and $Z_{in} = 73 + 42,5j = 84,47 < 30,2^\circ \Omega$ Hence the maximum voltage is at l_{emax}

$$V = 84,47 \left(\frac{10^{-3}}{\lambda} \cdot 5,06 \lambda \right) < 30,2^\circ = 0,43 < 30,2^\circ V$$

d) Effective Area

$$P_{rad} = R \left(\iint \frac{E \times H^*}{2} \cdot ds \right) = n \frac{\pi}{3} \left| \frac{I_0 l}{\lambda} \right|^2$$

$$U = \frac{r^2}{2n} |E_\theta(r, \theta, \phi)|^2$$

$$U_{max} = U(\theta = \frac{\pi}{2}) = \frac{n}{2} \left(\frac{k I_0 l}{4\pi} \right)^2$$

Sabiendo que:

$$D_0 = 4\pi \frac{U_{max}}{P_{rad}} = \frac{3}{2}$$

La maxima apertura efectiva, es entonces

$$A_{em} = \frac{\lambda^2}{4\pi} D_0 = \frac{3\lambda^2}{8\pi}$$

5. A small dipole made of Perfect Electric Conductor (PEC), of length $l = \lambda/20$ and wire radius $a = \lambda/400$ is fed symmetrically, and it is used as a communications antenna at the lower end of the VHF band ($f=30$ MHz). The input reactance of the dipole is given by:

$$X_{in} = -j20 \frac{\ln(l/2a) - 1}{\tan(\pi l/\lambda)}$$

while the input resistance is given by:

$$R_{rad} = 20\pi^2 \left(\frac{l}{\lambda} \right)^2$$

Determine:

- a) Input impedance of the antenna. Is it inductive/capacitive?.

$$Z_{in} = X_{in} + R_{rad}$$

Reemplazando los valores de l,a :

$$X_{in} = -j20 \frac{\ln(10) - 1}{\tan(\pi/20)} = -167,20j$$

$$R_{rad} = 20\pi^2(1/20)^2 = 0,493$$

$$Z_{in} = -167,20j + 0,493\Omega$$

Como X es imaginaria negativa, en este caso es una reactancia capacitiva.

- b) Radiation efficiency (in percent) The Radiation efficiency is given by :

$$n = \frac{R_r}{R_r + R_l} \cdot 100$$

Como R_l es la resistencia radiada y R_l es la resistencia de pérdidas. En un dipolo corto:

$$R_l = \frac{L}{6\pi a} \sqrt{\frac{\pi f \mu}{\sigma_{PEC}}}$$

Como un conductor perfecto (PEC) se tiene una conductividad infinita entonces $R_l = 0$ y entonces la eficiencia es de un 100 %

- c) Capacitor (in Farads) or inductor (in Henrys) that must be connected in series with the dipole in order to resonate it.

$$j\omega L = j167,20$$

$$L = 167,20/(2\pi(30E6)) = 0,88\mu H$$

6. Repeat exercise 5b considering a dipole made of aluminum and loss due to the skin effect.

$$\sigma_{Al(20^\circ)} = 3,5 \cdot 10^7$$

$$R_l = \frac{l}{6\pi a} \sqrt{\frac{\pi f \mu}{\sigma_{Al}}} =$$

$$R_l = 1,95E - 3\Omega$$

$$n = 100 \frac{0,493}{0,493 + 1,95E - 3} = 99,61 \%$$

7. This exercise deals with approximating a circular resonant loop with a 2-element array of resonant dipoles: one positioned symmetrically around the feed point and the other at the antipodal location.

- a) Compute the radiation pattern

$$V = \eta \frac{|I_o|^2}{8\pi^2} \left(\frac{\cos(\frac{kl}{2}\cos\theta) - \cos(\frac{kl}{2})}{\sin\theta} \right)^2$$

Como $V = B_o F$, el patrón de radiación es:

$$F(\theta) = \left(\frac{\cos(\frac{kl}{2}\cos\theta) - \cos(\frac{kl}{2})}{\sin\theta} \right)^2$$

para un diseño con $l = \frac{\lambda}{2}$, se tiene: $k = \frac{2\pi}{\lambda} = \frac{\pi}{\lambda}$

$$F(\theta) = \left(\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta} \right)^2$$

- b) Use numerical integration to compute antenna gain and radiation resistance
c) Compare these results to those obtained from fullwave simulation